

FULL-STACK AI & CAREER PLATFORM

# SkillForge API Engine

AI-Powered Skill Gap Analysis, Dynamic Career Roadmaps &  
RAG Guidance

AUTHOR & DEVELOPER

**MehmoodCoder**

ARCHITECTURE

**FastAPI + React 19 + Gemini AI**

SECURITY

**JWT Bearer & Role Control**

# Project Background & Core Objectives

SKILLFORGE OVERVIEW

## The Industry Problem

- **Misaligned Skill Sets:** Candidates struggle to evaluate their proficiency against fast-evolving tech standards (RAG, Vector DBs, Cloud).
- **Static & Generic Content:** Generic roadmaps lack personalized gap detection and project alignment.
- **AI Hallucination Risk:** Standard LLMs give unverified career guidance without company/domain grounding.

## The SkillForge Solution

- **Automated Gap Scoring:** Real-time quantitative skill gap evaluation comparing current vs target roles.
- **AI Roadmap Generation:** Custom structured milestones, recommended real-world projects, and curated learning paths.
- **Grounded RAG Assistant:** Domain-specific assistant backed by enterprise knowledge bases via Google Gemini API.

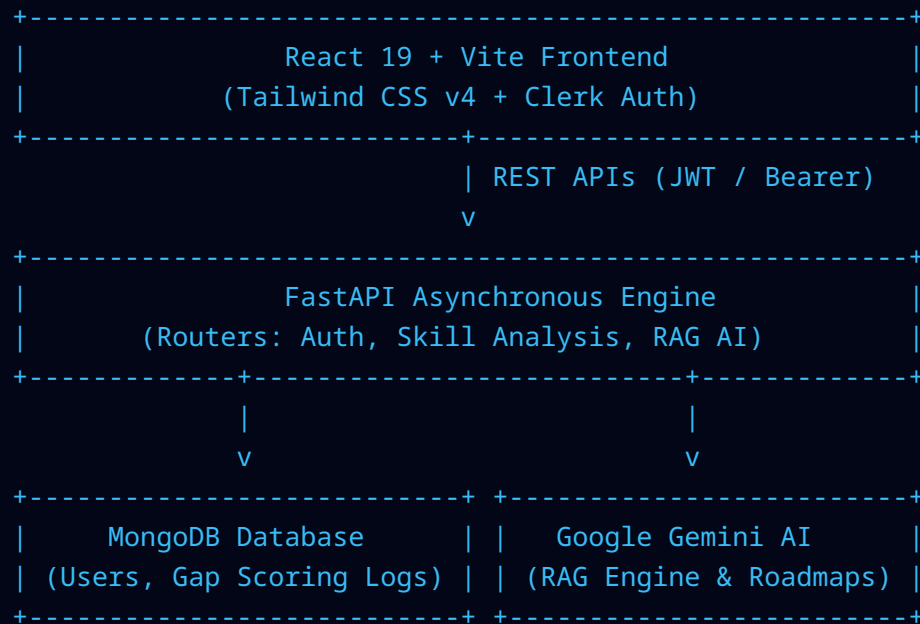
## Key Target Audience

Designed for **Students & Job Seekers** needing targeted career direction, **Mentors** reviewing competency scores, and **Enterprise Organizations** conducting upskilling programs.

# System Architecture & Ecosystem

## TECHNICAL DESIGN

### High-Level System Workflow



### Stack Breakdown

- **FastAPI** Async Python framework with automatic Swagger documentation.
- **React 19** High-performance UI built using Vite and Tailwind CSS v4.
- **MongoDB** Asynchronous database operations using **Motor** driver.
- **Google Gemini** Generative AI for structured roadmaps & grounded assistant.

# Backend Architecture & Security

## MODULES & DATA FLOW

| Module                  | Technical Implementation      | Functionality & Business Logic                                                                                   |
|-------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------|
| `app/core/security.py`  | OAuth2 + JWT + Bcrypt         | Handles password hashing and generates signed JWT tokens containing role scopes (Student, Mentor, Admin).        |
| `app/api/v1/auth.py`    | FastAPI Router (`/login`)     | Validates user credentials against MongoDB and issues secure Bearer authorization tokens.                        |
| `app/api/v1/student.py` | Gap Analysis & Roadmaps       | Computes missing competencies, calculates readiness percentage, and generates step-by-step career milestones.    |
| `app/api/v1/ai.py`      | RAG Engine (`/rag-assistant`) | Interfaces with Google Gemini AI and SkillForge vector indices to answer domain questions without hallucination. |



### Enterprise Security Features

Includes complete CORS middleware setup, JWT signature verification on all protected endpoints, and role-based access restrictions across API layers.

# API Contract & Core Endpoints

API REFERENCE

| Method | Endpoint                                        | Auth Required | Description & Payload Responsibilities                                           |
|--------|-------------------------------------------------|---------------|----------------------------------------------------------------------------------|
| POST   | <code>`/api/v1/auth/login`</code>               | None          | Authenticates user credentials and returns JWT Bearer token.                     |
| POST   | <code>`/api/v1/student/analyze-skills`</code>   | Bearer Token  | Accepts user skill inventory; returns percentage score and missing competencies. |
| POST   | <code>`/api/v1/student/generate-roadmap`</code> | Bearer Token  | Produces milestone timeline, project ideas, and curated learning links.          |
| POST   | <code>`/api/v1/ai/rag-assistant`</code>         | Bearer Token  | Executes grounded AI queries using SkillForge domain knowledge bases.            |

## Fully Tested & Interactive Documentation

All endpoints are verified and interactive via automatic OpenAPI / Swagger UI generated at <http://127.0.0.1:3000/docs>.

# Impact & Future Enhancements

## PROJECT ROADMAP

### 🌟 Current Accomplishments

- **Complete End-to-End Core:** Modular FastAPI backend connected seamlessly to React 19 UI.
- **Grounded AI Integration:** Reliable RAG-powered guidance eliminating generic responses.
- **Production Readiness:** Structured environment configuration (`.env`), async DB drivers, and clean security protocols.

### 🚀 Next Scalability Phase

- **Advanced Vector Stores:** Migration to FAISS / Qdrant cluster for scalable enterprise RAG indexing.
- **Automated Portfolio Scanning:** Direct integration with GitHub API to extract student commit skills automatically.
- **Live AI Mock Interviews:** Interactive voice/text assessment module for technical interview prep.

**Thank You!**

Developer Portfolio: <https://mh56-portfolio.vercel.app> | GitHub: <https://github.com/MehmoodCoder>